

Design Write-up

Architecture

The system replays one merged, time-ordered stream of ~52k quotes and 500 orders. A quote sharing a timestamp with an order is applied first, so every decision uses the most recent quote at or before “now”. The engine holds only the single current quote and never sees the tape, which makes lookahead impossible by construction. All prices are integer cents; no float comparisons exist anywhere.

Five small modules split the concerns: **data** (parsing only; the input files wrap each line in quotes, and nothing downstream knows), **book** (resting client limits in price-time priority, a lazily-pruned heap per side), **strategy** (every pricing and risk decision, as pure functions of order, position, time, and NBBO), **engine** (execution mechanics, position, cash), and **reporting**. All I/O paths and risk parameters live in `config.py`.

Two choices carry the compliance story. Every client fill passes through a single chokepoint, `engine._fill()`, which checks NBBO bounds and the client’s limit before booking. Decisions and constraints are separated, so a strategy bug crashes the replay instead of printing a violating fill. This covers the classic trap where the market moves through a resting limit: the order re-prices to the new touch or better, never its stale limit, and `test_engine.py` pins both directions. The second is `validate.py`, an independent audit that re-reads the raw CSVs and outputs, re-verifies every fill against the quote tape, reconstructs the firm’s position to zero, and recomputes the P&L.

Internalization strategy

The waterfall for every order, and for resting orders re-evaluated on every tick, is cross, then internalize, then route. Crosses are riskless and improve both sides, so they always win when limits overlap inside the NBBO. They are rare (1,600 shares) because marketable flow executes immediately rather than being held back to manufacture matches; holding a client’s order to fish for a cross trades their execution quality for the firm’s.

Pricing. Principal fills price at the midpoint rounded a half-cent in the firm's favor; at a 2¢ spread this is the exact midpoint. The rule that an internalized client always does strictly better than routing is a self-imposed best-execution principle (the prompt requires only at-or-within NBBO), and with whole-cent prices it makes 2¢ the minimum feasible spread for new risk. The principle costs nothing: at 1¢ the firm's half-cent edge at the touch equals its hedge cost, so that flow is zero-expectation anyway. Two exceptions fill at other prices. Inventory-reducing flow at a 1¢ spread fills at the touch, where the client matches routing and the firm exits without paying the spread. The rebalancer fills resting inside-spread limits at the client's own price, a cheaper exit than hedging at the touch and a fill the client wasn't otherwise getting.

Risk. A hard limit of $\pm 10,000$ shares is enforced by sizing: every principal fill is truncated to remaining capacity and the residual routes, so a breach cannot happen. A soft band of $\pm 6,000$ triggers immediate reduction, resting orders first, market second. From 15:55 the firm is reduce-only; at 16:00 it flattens. This day it was already flat by 13:45 through client flow, and the 17 leftover DAY orders expired. Routing a non-marketable limit would violate the client's price, so expiry is the only compliant disposition. On sizing: 10k shares is about 0.02% of AAPL's real-world ADV and 1.6% of typical last-five-minutes volume, so the forced flatten is always liquid. The limit itself comes from the risk budget rather than liquidity.

What is optimized, and the honest accounting. The objective is spread capture net of hedging costs, subject to never disadvantaging a client versus routing and to hard risk bounds. Attribution against the contemporaneous mid decomposes the day's +\$14,396 exactly: +\$160 of execution edge (+\$545 from client fills through odd-spread rounding and touch exits, minus \$385 of hedge-crossing costs at 1.8¢/share) and +\$14,236 of inventory drift, \$11.4k of it in the 11:00 selloff hour when the book happened to be short. Midpoint pricing hands the client nearly the whole spread, so the realized profit is the position sitting on the right side of an engineered day. That is luck. The band only bounds it. The same trap runs the other way: wide spreads make internalizing look most attractive exactly when the market is trending and the exit costs half of that same wide spread. The break-even condition is explicit: internalize only if expected capture from future offsetting flow exceeds the probability-weighted hedge cost plus the expected adverse move while holding.

Trade-offs accepted. No trend prediction, since fitting a signal to one simulated day is curve-fitting; the band bounds the damage instead. Static thresholds and limits. The 1¢ new-risk flow forgone. One refinement was tested and rejected: unwinding 500 shares whenever the spread is 1¢ and the book sits pinned at the soft limit fired only 4 times all day, because pinned inventory and tight spreads anti-correlate. That is direct evidence the hedging problem is procyclical and has to be solved before the book is pinned.

Assumptions

Routed orders and firm hedges fill fully at the touch regardless of displayed size (per the prompt). Multiple quotes within a millisecond apply in file order; each fill records the exact quote it executed against. The close uses the last quote of the session as the prevailing NBBO.

Production changes

Symbols and feeds: a feed handler per symbol pushing into per-symbol engine instances; the engine already consumes only an event stream, so the replay harness is the disposable part. Sequence numbers and staleness detection on quotes, since a stale NBBO silently invalidates “compliant” fills. **Risk:** dollar limits, volatility-targeted soft bands with hysteresis (constant dollar risk instead of constant shares; the hedge blotter’s `position_after` column pinned at $\pm 6,000$ shows the static band degenerating into a full-band directional position during trends), and position limits that shrink with time remaining: $\text{capacity} \approx \text{ADV} \times (\text{remaining} / 390) \times \text{participation rate (5–10\%)}$. At 5 minutes and 5% participation that formula gives $\sim 32,000$ shares, so the hard-coded 10,000 is conservative by 3×. MOC orders for the flatten, and a kill switch to route-everything. **Hedge execution:** the model assumes hedges fill fully at the touch; in production the firm chooses between crossing the spread and posting inside it. Compare a vol-based fair-value estimate against the touch and cross only when the gap exceeds a threshold, working the rest passively. **Pricing:** sub-penny price improvement (making 1¢ spreads internalizable); inventory-risk-adjusted principal pricing that skews the fill price away from firm-favorable midpoint as the book fills up; and per-client markout tiers, a toxicity multiplier on the 2¢ spread threshold per client. Treating all flow as equally benign is the biggest realism gap. **Costs:** internalize-vs-route decisions should net out exchange fees and market impact, both ignored here. **Data:** with full tick

and depth data instead of L1 snapshots, routed fills could walk the book realistically rather than filling at the displayed touch (unneeded here by the prompt's assumptions).

Plumbing: explicit OMS order-state machine with client execution reports, journaled fills and position for intraday restart, and the `validate.py` audit run post-trade as an independent compliance process.